Jeremy (Jianming) Zheng

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Education

University of California, Irvine — Irvine, CA
B.S. in Computer Science and Engineering | Magna Cum Laude, Phi Beta Kappa — *Sep. 2022 – Mar. 2026*

- **Awards**: Dean's Honors List (10x), Hackathon Winner (1x)

Georgia Institute of Technology — Atlanta, GA
M.S. in Computational Science and Engineering — *Jan. 2027 – May. 2028*

Experience

Undergraduate Research Assistant — Jan. 2025 – Feb. 2026
UCI Embedded and Cyber-Physical Systems Lab — *Irvine, CA*

- Enabled controlled testing of DJI Tello perception under degraded vision by developing Python tools to inject blur and distortion into live video feeds.
- Identified capture, processing, and inference bottlenecks in perception pipelines operating at 10–40 FPS by profiling end-to-end performance on resource-constrained edge hardware.
- Evaluated localization drift and loop closure behavior under dynamic conditions by deploying ORB-SLAM3 within ROS.

Projects

AuraBot: An Edge AI Desk Wellness Companion | *Software Lead* — Sep. 2025 – Mar. 2026

- Enabled concurrent sensing, communication, and device control on ESP32/ESP32-P4 by implementing C firmware with ESP-IDF, FreeRTOS tasks, interrupt handling, and event-driven state machines.
- Achieved approximately 95% recognition accuracy for timer-related voice commands during live operation by integrating ESP32 I2S audio capture with Raspberry Pi speech processing.
- Resolved an Opus audio-task stack overflow by increasing task stack allocation, moving large buffers off the stack, and placing suitable voice buffers in PSRAM.
- Enabled communication across embedded subsystems by developing and validating I2C, SPI, PWM, and Wi-Fi interfaces under concurrent workloads.

Bincredible (BRDG Innovative Challenge) | *Software Lead* — May 2025 – Oct. 2025

- Delivered an integrated trash-sorting prototype with a six-person team by leading software development, model deployment, and hardware/software integration.
- Achieved 93% trash-classification accuracy by training a YOLOv11 classifier and deploying TensorFlow Lite inference on Raspberry Pi 5.
- Reduced Firebase query time by 25% by implementing database indexes to optimize backend queries.

Smart-Sit Guardian | *Embedded Software Engineer* — Sep 2024 – Dec. 2024

- Enabled sensor acquisition at three-second intervals by implementing ESP32 firmware in C++ to sample multiple sensors over I2C.
- Resolved sensor communication and timing issues by tracing I2C acquisition behavior through UART logs.
- Enabled cloud collection of ESP32 sensor readings by building an MQTT data pipeline with AWS IoT Core.

Technical Skills

Programming: C, C++, Rust, Python
Embedded & Systems: Embedded C, ESP32 (ESP-IDF), FreeRTOS, Real-Time Systems, Interrupts, Device Drivers, Memory Management, DMA
Hardware Interfaces: I2C, SPI, UART, PWM, Wi-Fi
Systems & Networking: Embedded Linux, Multithreading, TCP/IP, MQTT, WebSocket
AI / Edge ML: TensorFlow Lite, PyTorch, YOLO, OpenCV
Tools & Debugging: Git, GitHub, CMake, Make, UART Serial Logging, Logic Analyzer, Oscilloscope, VS Code
Languages: Mandarin (Fluent)